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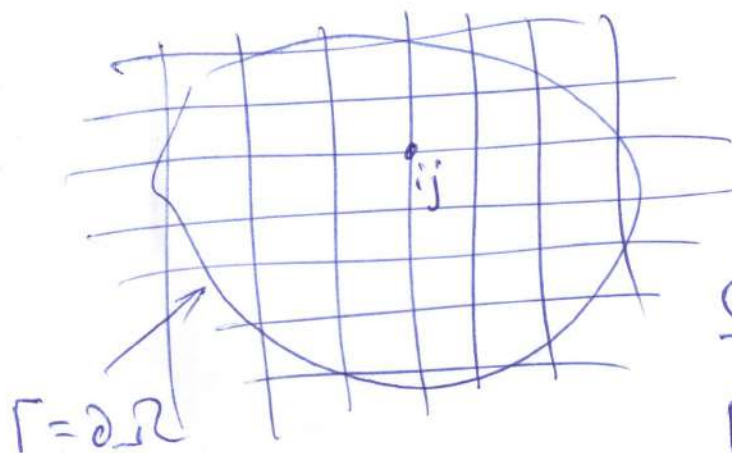
N₂ 2

11.09.2024

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f(x, y)$$



$$u|_{\partial\Omega} = \varphi(x, y)$$



$$x_i = x_0 + i h_1$$

$$y_j = y_0 + j h_2$$

$$\Omega_h = \{(i, j) \mid (x_i, y_j) \in \Omega\}$$

$$\Gamma_h = \{(i, j) \mid (x_i, y_j) \in \Gamma\}$$

MK P

$$\frac{1}{h_1^2}(u_{i+1,j} - 2u_{i,j} + u_{i-1,j}) + \frac{1}{h_2^2}(u_{i,j+1} - 2u_{i,j} + u_{i,j-1}) = f_{i,j} \quad (i, j) \in \Omega_h$$

$$u_{i,j} = \varphi_{i,j} \quad \text{on } \Gamma_h$$

B-P M

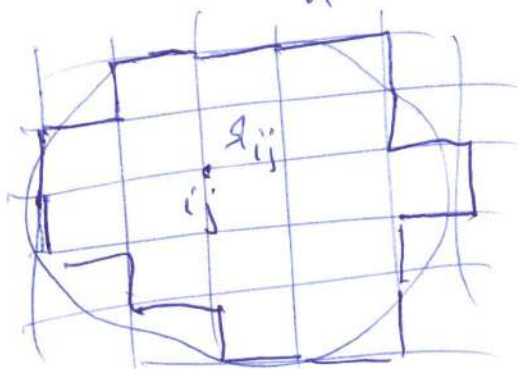
continuous →

$$W = \frac{1}{2} \int_{\Omega} \left(\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 \right) dx dy \rightarrow \min$$

$$u|_{\partial\Omega} = \varphi(x, y)$$

$$W \rightarrow W_h$$

$$W_h = \frac{1}{2} \sum_{(i,j) \in \Omega_h} \int_{Q_{ij}} \left(\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 \right) dx dy$$



$$Q_{ij} = \text{rectangle with vertices } (i, j), (i+1, j), (i+1, j+1), (i, j+1)$$

[2]

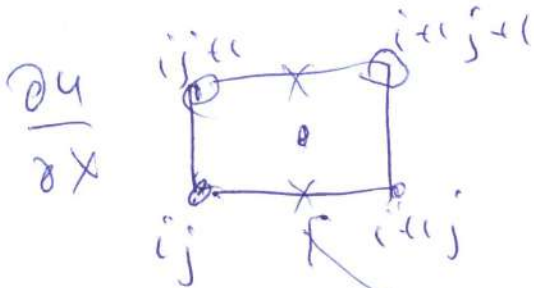
q_{ij}



$$\int_{q_{ij}} f(x,y) dx dy \approx f(x_{ij}, y_{ij}) \cdot \Delta S'$$

q_{ij}

ΔS - площадь элемента



$$\frac{\partial u}{\partial x} \approx \frac{1}{2} \left(\frac{1}{h_1} (u_{i+1,j} - u_{i,j}) + \frac{1}{h_1} (u_{i+1,j+1} - u_{i,j+1}) \right)$$

$$\frac{\partial u}{\partial x} \approx \frac{1}{2h_1} (u_{i+1,j} + u_{i+1,j+1} - \underline{u_{i,j}} - u_{i,j+1}) = (d_1^+ u)_{ij}$$

$$\frac{\partial u}{\partial y} \approx \frac{1}{2h_2} (u_{i,j+1} + u_{i+1,j+1} - \underline{u_{i,j}} - u_{i+1,j}) = (d_2^+ u)_{ij}$$

$$\int_{q_{ij}} (d_1^+ u)_{ij}^2 + (d_2^+ u)_{ij}^2 dx dy = \Delta S' [(d_1^+ u)_{ij}^2 + (d_2^+ u)_{ij}^2]$$

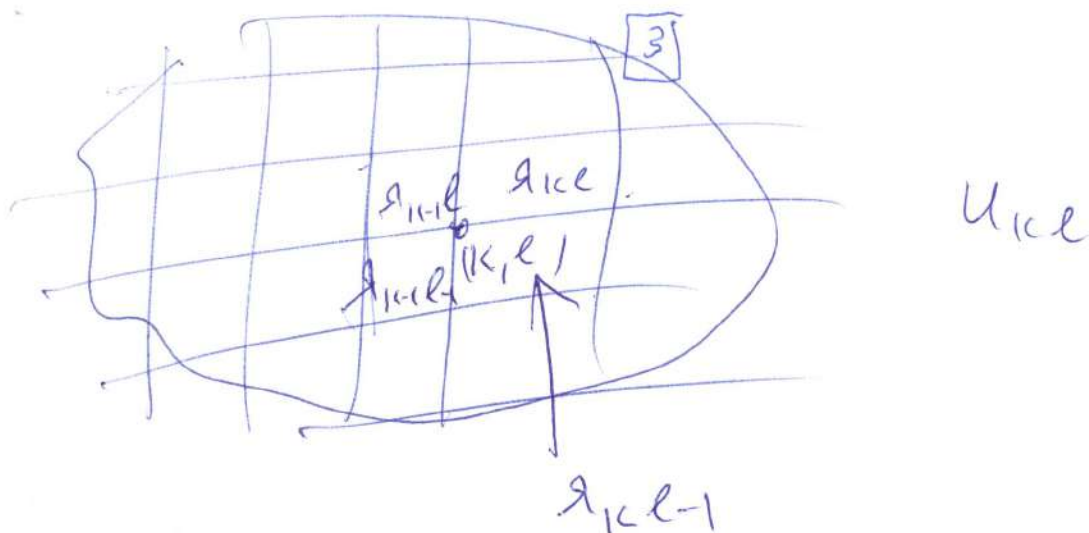
$$W_h = \frac{\Delta S'}{2} \sum_{(i,j) \in \mathcal{R}_h} [(d_1^+ u)_{ij}^2 + (d_2^+ u)_{ij}^2] \quad \leftarrow \text{дискретизация}$$

сеточные функции

$$W_h = W_{h \min}, \text{ если } \frac{\partial W_h}{\partial u_{k,l}} = 0 \quad (k,l) \in \mathcal{R}_h$$

$$W_h = \frac{\Delta S}{2} \sum_{(i,j) \in \mathcal{R}_h} F_{ij}$$

$$F_{ij} = [(d_1^+ u)_{ij}^2 + (d_2^+ u)_{ij}^2]$$



$$\frac{\partial W_h}{\partial u_{ke}} = \frac{\partial}{\partial u_{ke}} \left[\frac{\Delta S'}{2} (F_{ke} + F_{k-1,e} + F_{k-1,l-1} + F_{k,l-1}) \right] =$$

$$\textcircled{\textcircled{I} \textcircled{II}} = \frac{\Delta S}{2} \frac{\partial}{\partial u_{1k\ell}} (F_{1k} + F_{1k-1\ell} + F_{1k-1\ell-1} + F_{1k\ell-1}) = 0$$

$$\frac{\partial F_{ice}}{\partial u_{ice}} = \frac{\partial}{\partial u_{ice}} \left[(d_1^T u)_{ice} + \frac{1}{2} (d_2^T u)_{ice}^2 \right] =$$

$$= \frac{1}{2} \left[\lambda (d_1^+ u)_{ice} \frac{\partial}{\partial u_{ice}} (d_1^+ u)_{ice}^2 + \lambda (d_2^+ u)_{ice} \frac{\partial}{\partial u_{ice}} (d_2^+ u) \right] =$$

$$= (d_1^+ u)_{ice} \left(-\frac{1}{2u_1} \right) + (d_2^+ u)_{ice} \left(-\frac{1}{2u_2} \right)$$

$$\frac{\partial F_{ice}}{\partial u_{ice}} = -\frac{1}{2h_1} (d_1^+ u)_{ice} - \frac{1}{2h_2} (d_2^+ u)_{ice}$$

$$\frac{\partial F_{k-1}}{\partial u_{k-1}} = \frac{1}{2h_1} (d_1^T u)_{k-1} e - \frac{1}{2h_2} (d_2^T u)_{k-1} e$$

$$\frac{\partial F_{k-1, l-1}}{\partial u_{k, l}} = \frac{1}{2h_1} (d_1^T u)_{k-1, l-1} + \frac{1}{2h_2} (d_2^T u)_{k-1, l-1}$$

$$\frac{\partial F_{k,l-1}}{\partial u_{k,l}} = -\frac{1}{2h_1}(d_1^T u)_{k,l-1} + \frac{1}{2h_1}(d_2^T u)_{k,l-1}$$

جواباً } (*)
توضیحاً
4
x

$d_{k\ell}^0$, $\hat{c}_{k\ell}$ da $\hat{p}_{k\ell}$ $\hat{r}_{k\ell}$
 $d_{k\ell}^+ = \frac{1}{2} \left(\hat{c}_{k\ell} + \hat{p}_{k\ell} \right)$
 $d_{k\ell}^- = \frac{1}{2} \left(\hat{c}_{k\ell} - \hat{p}_{k\ell} \right)$

→ up 4

(4)

$$(d_1^+ u)_{ke} = \frac{1}{2h_1} (u_{k+1,e} + u_{k+1,e-1} - \underline{u_{ke}} - u_{ke+1})$$

$$(d_1^+ u)_{k-1,e} = \frac{1}{2h_1} (\underline{u_{ke}} + u_{ke+1} - u_{k-1,e} - u_{k-1,e+1})$$

$$(d_1^+ u)_{k-1,e-1} = \frac{1}{2h_1} (u_{ke-1} + \underline{u_{ke}} - u_{k-1,e-1} - u_{k-1,e})$$

$$(d_1^+ u)_{k,e-1} = \frac{1}{2h_1} (u_{k+1,e-1} + u_{k+1,e} - \underline{u_{ke-1}} - \underline{u_{ke}})$$

$$(d_2^+ u)_{ke} = \frac{1}{2h_2} (u_{k,e+1} + u_{k+1,e+1} - \underline{u_{ke}} - u_{k+1,e})$$

$$(d_2^+ u)_{k-1,e} = \frac{1}{2h_2} (u_{k-1,e+1} + u_{k,e+1} - u_{k-1,e} - \underline{u_{ke}})$$

$$(d_2^+ u)_{k-1,e-1} = \frac{1}{2h_2} (u_{k-1,e} + \underline{u_{ke}} - u_{k-1,e-1} - u_{ke-1})$$

$$(d_2^+ u)_{k,e-1} = \frac{1}{2h_2} (\underline{u_{ke}} + u_{k+1,e} - u_{ke-1} - u_{k+1,e-1})$$

$$\begin{aligned} \frac{\partial W_h}{\partial u_{ke}} &= \frac{\Delta S}{2h_1} [-d_1^+ u_{ke} + d_1^+ u_{k-1,e} + d_1^+ u_{k-1,e-1} - d_1^+ u_{k,e-1}] + \\ &+ \frac{\Delta S}{2h_2} [-(d_2^+ u)_{ke} - d_2^+ u_{k-1,e} + d_2^+ u_{k-1,e-1} + d_2^+ u_{k,e-1}] = \end{aligned}$$

[5]

$$\frac{\partial w_h}{\partial u_{k\ell}} = \frac{\Delta S'}{4h_1^2} \left[-u_{k-1,\ell+1} + 2u_{k,\ell+1} - u_{k+1,\ell+1} + 2u_{k,\ell} + 4u_{k,\ell} - 2u_{k,\ell-1} \right. \\ \left. - u_{k-1,\ell-1} + 2u_{k,\ell-1} - u_{k+1,\ell-1} \right] + \frac{\Delta S'}{4h_2^2} \left[\right]$$

$$\frac{\Delta S'}{4h_1^2}$$

-1	2	-1
-2	4	-2
-1	2	-1

$$\frac{1}{4h_1^2}$$

1	-2	1
2	-4	2
1	-2	1

$$\frac{1}{h_1^2} (u_{i-1,j} - 2u_{i,j} + u_{i+1,j}) \sim \frac{\partial^2 u}{\partial x^2}$$

$$\frac{1}{2} \left[\frac{1}{2} \begin{array}{|c|c|c|} \hline 1 & -2 & 1 \\ \hline 1 & -4 & 1 \\ \hline 1 & -2 & 1 \\ \hline \end{array} + \begin{array}{|c|c|c|} \hline 1 & -2 & 1 \\ \hline 1 & -4 & 1 \\ \hline 1 & -2 & 1 \\ \hline \end{array} \right]$$

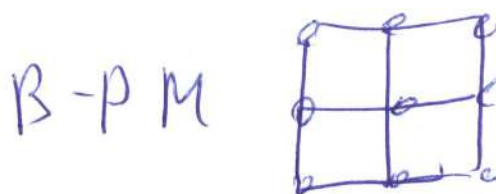
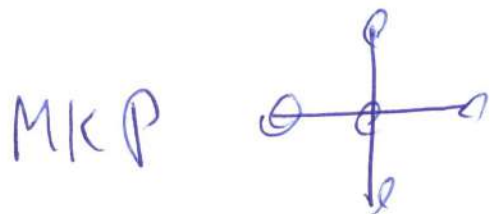
$$\frac{\partial w_h}{\partial u_{k\ell}} = \Delta S' \left[\frac{1}{4h_1^2} \begin{array}{|c|c|c|} \hline -1 & 2 & -1 \\ \hline -2 & 4 & -2 \\ \hline -1 & 2 & -1 \\ \hline \end{array} + \frac{1}{4h_2^2} \begin{array}{|c|c|c|} \hline -1 & -2 & -1 \\ \hline 2 & 4 & 2 \\ \hline -1 & -2 & -1 \\ \hline \end{array} \right]$$

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$$MKP \quad \frac{1}{h_1^2}(u_{i+1,j} - 2u_{i,j} + u_{i-1,j}) + \frac{1}{h_2^2}(u_{i,j+1} - 2u_{i,j} + u_{i,j-1}) = G_{ij}$$

B-P M

$$\frac{1}{4h_1^2} [u_{i+1,j+1} - 2u_{i,j+1} + u_{i-1,j+1}] + \frac{1}{4h_1^2} [u_{i+1,j} - 4u_{i,j} + u_{i-1,j} + u_{i+1,j-1} - 2u_{i,j-1} + u_{i-1,j-1}] + \frac{1}{4h_2^2} [u_{i,j+1} - 2u_{i,j} + u_{i,j-1} + 2u_{i,j+1} - 4u_{i,j} + 2u_{i,j-1} + u_{i-1,j+1} - 2u_{i-1,j} + u_{i-1,j-1}] = 0$$



u_p - e функция

(*) $W = \frac{1}{2} \int_{\Omega} \left(\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 \right) dx dy$

$W \rightarrow \min_{p_1, p_2}$

u_p - e функция (x, x)

$W = \frac{1}{2} \int_{\Omega} \left(\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 \right) dx dy$

$W = \int_{\Omega} F(u, p_1, p_2) d\Omega$

$p_1 = \frac{\partial u}{\partial x}, p_2 = \frac{\partial u}{\partial y}$

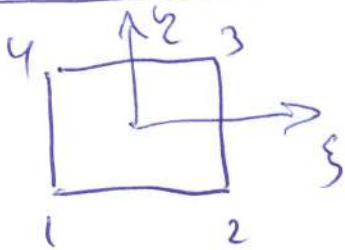
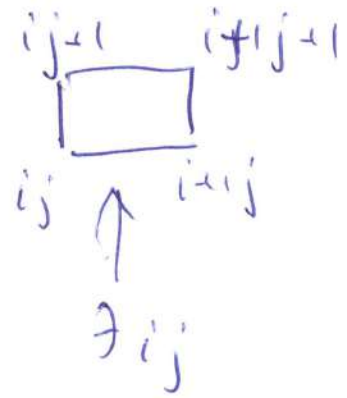
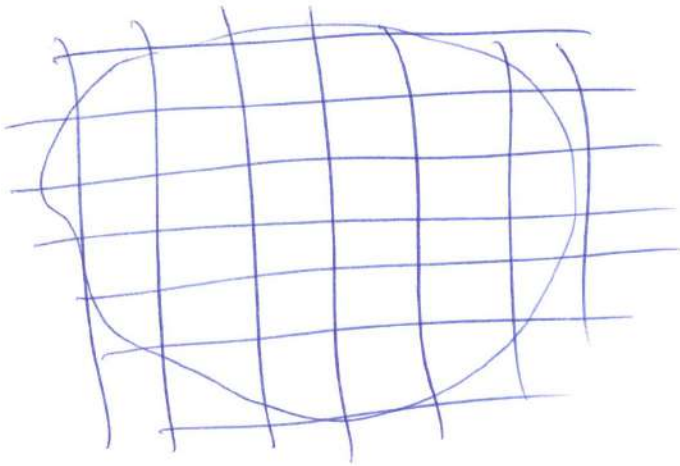
u_p - e функция: $\left[\frac{\partial F}{\partial u} - \frac{\partial}{\partial x} \frac{\partial F}{\partial p_1} - \frac{\partial}{\partial y} \frac{\partial F}{\partial p_2} = 0 \right]$

(*) $\rightarrow \frac{1}{2} \left[\frac{\partial}{\partial x} \left(2 \frac{\partial u}{\partial x} \right) - \frac{\partial}{\partial y} \left(2 \frac{\partial u}{\partial y} \right) \right] = - \frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} = 0$

(**) $\frac{1}{2} \cdot 2 f - \frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} = 0 \Rightarrow \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f(x, y)$

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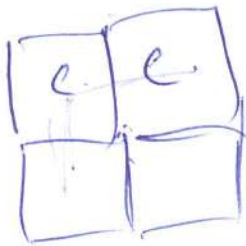
MKJ



$$u = c_0 + c_1 x + c_2 y + c_3 xy$$

$$u = \sum_{i=1}^4 c_i \psi_i(x, y)$$

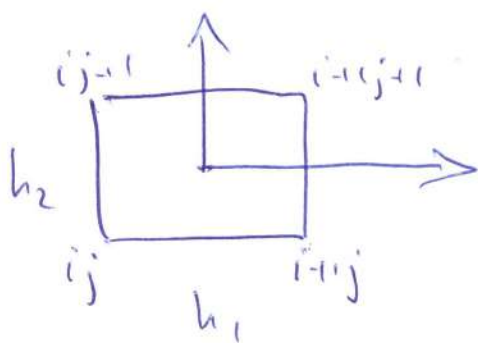
$$W \rightarrow W_h = \sum_{(i,j) \in \Omega_h} \int_{\partial i_j} F(u(c_1, \dots, c_n)) d\Omega$$



$$\Rightarrow \frac{\partial F}{\partial (c_n)_{ij}} = 0$$

$$c_i = c(u_1, u_2, u_3, u_4)$$

$$\Rightarrow \frac{\partial F}{\partial u_{ice}} = 0$$



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$$(i, j) \rightarrow \left(-\frac{h_1}{2}, -\frac{h_2}{2}\right)$$

$$(i-1, j) \rightarrow \left(\frac{h_1}{2}, -\frac{h_2}{2}\right)$$

$$(i+1, j-1) \rightarrow \left(\frac{h_1}{2}, \frac{h_2}{2}\right)$$

$$(i, j+1) \rightarrow \left(-\frac{h_1}{2}, \frac{h_2}{2}\right)$$

$$u = c_0 + c_1 x + c_2 y + c_3 xy$$

$$u_{ij} = c_0 - c_1 \frac{h_1}{2} - c_2 \frac{h_2}{2} + c_3 \frac{h_1 h_2}{4}$$

$$u_{i+1,j} = c_0 + c_1 \frac{h_1}{2} - c_2 \frac{h_2}{2} - c_3 \frac{h_1 h_2}{4}$$

$$u_{i,j+1} = c_0 - c_1 \frac{h_1}{2} + c_2 \frac{h_2}{2} - c_3 \frac{h_1 h_2}{4}$$

$$u_{i+1,j+1} = c_0 + c_1 \frac{h_1}{2} + c_2 \frac{h_2}{2} + c_3 \frac{h_1 h_2}{4}$$

$$d_3^+ =$$

$$= \frac{1}{h_1 h_2} \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$c_0 = \frac{1}{4} [u_{ij} + u_{i+1,j} + u_{i,j+1} + u_{i+1,j+1}] = (d_0^+ u)_{ij} \sim \frac{\partial^2}{\partial x^2}$$

$$c_1 = \frac{1}{2h_1} [u_{i+1,j} + u_{i+1,j+1} - u_{ij} - u_{i,j+1}] = (d_1^+ u)_{ij} \sim \frac{\partial}{\partial x}$$

$$c_2 = \frac{1}{2h_2} [u_{i,j+1} + u_{i+1,j+1} - u_{ij} - u_{i+1,j}] = (d_2^+ u)_{ij} \sim \frac{\partial}{\partial y}$$

$$c_3 = \frac{1}{h_1 h_2} [u_{ij} + u_{i+1,j+1} - u_{i+1,j} - u_{i,j+1}] = (d_3^+ u)_{ij} \sim \frac{\partial^2}{\partial x \partial y}$$

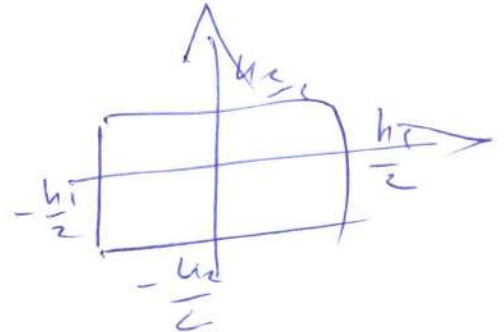
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$$W = \int_{\Omega} \left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 dx dy$$

$$u = c_0 + c_1 x + c_2 y + c_3 xy$$

$$\frac{\partial u}{\partial x} = c_1 + c_3 y$$

$$\frac{\partial u}{\partial y} = c_2 + c_3 x$$



$$W_h = \sum_{(i,j) \in \Omega_h} \int_{\partial ij} (c_1 + c_3 y)^2 + (c_2 + c_3 x)^2 dx dy$$

$$\int_{\partial ij} \left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 dx dy = \int_{-\frac{h_2}{2}}^{\frac{h_2}{2}} \left[\int_{-\frac{h_1}{2}}^{\frac{h_1}{2}} (c_1 + c_3 y)^2 + (c_2 + c_3 x)^2 dx \right] dy$$

$$= \int_{-\frac{h_2}{2}}^{\frac{h_2}{2}} \int_{-\frac{h_1}{2}}^{\frac{h_1}{2}} c_1^2 + 2c_1 c_3 y + c_3^2 y^2 + c_2^2 + 2c_2 c_3 x + c_3^2 x^2 dx dy =$$

$$= h_1 h_2 (c_1^2 + c_2^2) + c_3^2 \frac{h_1 h_2^3}{12} + c_3 \frac{h_1^3 h_2}{12} =$$

$$= (c_1^2 + c_2^2) h_1 h_2 + c_3^2 h_1 h_2 \left(\frac{h_1^2 + h_2^2}{12} \right)$$

$$W_h = \sum_{ij} h_1 h_2 \sum_{(i,j) \in \Omega_h} \left[(d_1^T u)^2 + (d_2^T u)^2 + \frac{h_1^2 + h_2^2}{12} (d_3^T u)^2 \right]$$

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$$\begin{pmatrix} x_{ij} \\ y_{ij} \end{pmatrix} = B_n \begin{pmatrix} i \\ j \end{pmatrix} - \text{решетка Браве}$$

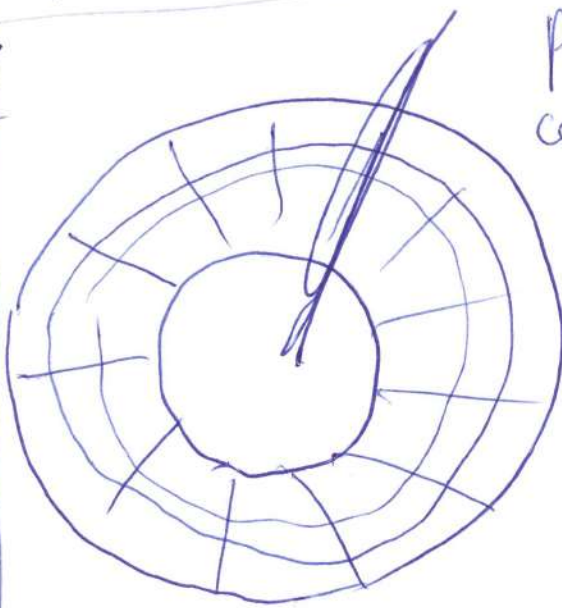
$\in R^n : (i) = (i_1, i_2, \dots, i_n)$ - мультииндекс

$$(1) \bar{X}_{(i)} = B_n \cdot (i) + \bar{X}_0$$

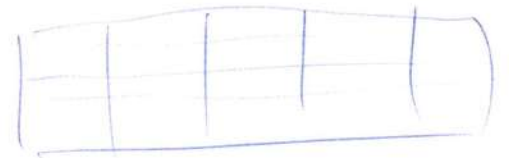
R^n , (i) - n-мерный индекс

Код сетки

1	4	3	2
2	3	7	6
3	8	11	7
3	4	9	8
4	5	6	9
6	7	12	11
8	8	13	12
9	14	18	13
9	16	15	14
12	17	16	11
12	13	18	17



пер. $\mathcal{Y}_{(i)}$ $\mathcal{Y}_{i,j} = [(i,j), (i+1,j+1)]$
сетка $(i,j+1), (i+1,j+1)$



куб PANSYS

